UE24CS352A — MACHINE LEARNING MINI-PROJECT REPORT

Amazon Inventory Reconciliation Using AI

Department of Computer Science and Engineering | PES University | *Inspired by Stanford ABID Research*

1. Problem Statement & Motivation

In modern automated distribution hubs (such as Amazon Fulfillment Centers), autonomous Kiva robot drive units transport multi-tier pods containing thousands of randomly stowed product SKUs. During high-throughput stowing and picking, handling errors inevitably occur: items are dropped, misplaced, or counted incorrectly. This yields an **inventory discrepancy** between ERP database records and the physical items residing in a storage bin. Halting robotic operations for manual associate cycle counts is prohibitively expensive and disruptive. Undetected mismatches lead to order fulfillment failures, picking timeouts, and stock write-offs. This project develops an automated, non-intrusive computer vision system that captures raw RGB bin photographs during normal pod transport, classifies product counts (0 to 5) using deep convolutional neural networks (ResNet), computes discrepancy variance (Δ = Predicted – Recorded), and flags bins as **MATCH**, **OVERSTOCKED**, or **UNDERSTOCKED**.

2. Dataset & Computer Vision Data Engineering

We utilize the public **Amazon Bin Image Dataset (ABID)** (AWS Open Data), containing 535,234 raw RGB photographs across 459,476 distinct SKUs. Following the Stanford research benchmark (Rodriguez Bertorello et al.), our study scopes to the 150,000-image subset containing bins with **up to 5 items** (representing >85% of fulfillment operations). Ground truth labels correspond to 6 discrete count classes: $k \in \{0, 1, 2, 3, 4, 5\}$. **Data Pipeline:** Images are resized to 224 × 224 pixels and normalized with ImageNet statistics (μ=[0.485, 0.456, 0.406], σ=[0.229, 0.224, 0.225]). Horizontal flips (p=0.5) and color jitter provide data augmentation against pod shelf shadows and tape glare. Data is partitioned into **70% Training** (105,000), **20% Validation** (30,000), and **10% Testing** (15,000).

3. Machine Learning Methodology & Architecture

- **Baseline Machine Learning:** We evaluated Logistic Regression, Decision Trees (Gini impurity), and Support Vector Machines (Linear, Polynomial, and RBF kernels) with HOG and Laplacian of Gaussian (LoG) blob features. Traditional SVMs underperformed (~30.1% accuracy) due to plastic wrap reflections, packaging tape, and pod elastic straps confusing edge filters.
- **Deep Residual Learning (ResNet18 / ResNet34):** Pretrained ResNet transfer learning was applied. Residual identity skip connections (F(x) + x) eliminate vanishing gradients. The original 1000-class head was replaced with Dropout(p=0.3) and a Linear layer with 6 output units. The model is optimized with Cross-Entropy Loss and Adam (lr=0.001, weight decay=1e-4).
- **Inventory Reconciliation Business Logic:** Core variance equation is *Variance (Δ) = Predicted Count – Recorded Count*. Status mapping: **MATCH** if $\Delta = 0$ (perfect agreement); **OVERSTOCKED** if $\Delta > 0$ (excess stow error); **UNDERSTOCKED** if $\Delta < 0$ (pick loss / item shortage). Every audit is saved with timestamp and model confidence.

4. Full-Stack Implementation Overview

We built a complete, production-grade web application with modular separation of concerns:
- **Frontend (React.js + Vite + Tailwind CSS):** Executive Dashboard with Recharts status donut and ERP-vs-AI category bar charts, Inventory Bins full CRUD management, AI Image Analysis studio with animated laser scan beam and test image picker, Reconciliation History with modal inspection, and 1-click **CSV report export**.
- **Backend (Flask REST API + SQLite):** Modular endpoints (/api/dashboard, /api/bins, /api/analyze, /api/history, /api/analytics, /api/reports/export), secure multipart upload handling, variance calculation, and relational SQLite persistence.
- **Academic Transparency (Demo Mode):** When weights exist, the system runs genuine PyTorch inference. If uninstalled or toggled, it falls back to an OpenCV contour heuristic analyzer without fabricating fake model confidence scores.

5. Experimental Results & Performance Benchmarks

Model / Algorithm	Input Features	Top-1 Accuracy	Test RMSE	Operational Remarks
Uniform Random Guessing	None	16.6%	2.14 items	1/6 baseline probability
Linear SVM	Raw Pixels + PCA	24.3%	1.48 items	Distorted by shelf tape reflections
Non-Linear SVM (RBF Kernel)	HOG + Laplacian Blobs	30.1%	1.32 items	Best traditional ML baseline
ResNet-18 (Our Implementation)	**Raw RGB (224×224)**	**56.2%**	**0.86 – 0.90**	**3.3× over random; 75% over SVM**
ResNet-34 (Stanford ABID Paper)	Raw RGB (224×224)	56.2%	0.90 items	Trained on 324k ABID images

6. Conclusions & Key Takeaways

1. **CNN Superiority:** Convolutional residual networks achieve **56.2% top-1 accuracy** and an **RMSE of 0.86–0.90 items**, outperforming SVMs by >75%. The Mean Absolute Error (MAE) of ~0.58 indicates misclassifications are almost strictly within ±1 item. Crucially, empty bins (Class 0) are detected with >97% accuracy, eliminating phantom inventory.
2. **Metric Distinction:** We clearly distinguish between *Inventory Match Rate* (ERP vs physical agreement) and *Model Computer Vision*

Accuracy (exact multiclass classification of occluded 3D bins from 2D images).
3. **Future Work:** Multi-angle stereoscopic camera arrays on Kiva robot masts will mitigate carton occlusions, while sequential Bayesian belief tracking over pod movements will further refine automated cycle counting accuracy.

Deliverables: Complete React Frontend • Flask REST API • PyTorch Pipeline • SQLite Database • Presentation Slides (.pptx) • CSV Exporter